Random Walk of a Photon from the Center of the Sun to outside

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1 Introduction

The path followed by a photon generated in the center of the Sun on its way out of the Sun is complicated by the fact that the photon interacts with many other particles along the way. Even though the speed of a photon is very high, the direction of its motion is changed with every interaction. This is easy to model in principle, but the amount of computation time it takes to model these repeated changes is enormous, so various techniques are applied to reduce the computational burden of this simulation. The approximate behavior of such a photon has been modeled in this project using python programming. The resulting programs have been uploaded to Github.

2 Theory and Equations

2.1 Motion of a photon over a single second

The motion of a photon within the sun can be expressed as a random walk, since the photons are, on average, equally likely to be scattered in all directions with each interaction. In practice, a photon changes direction about 10^8 times per second on its way out from the center of the Sun. The distance crossed by the photon between collisions, then, is given by

$$s = \frac{3^8}{10^8} meters \tag{1}$$

But this can be in any direction, so the x, y, z components of displacement are basically random numbers. For a large number of scatterings or a large number of photons, these numbers follow a gaussian distribution centered on zero with a standard deviation of 0.00577.

So we can model the motion of a photon in the sun by taking random values of x,y,z from a gaussian distribution with mean zero and standard deviation 0.00577. Each such set represents the displacement over a single step. We can model successive steps by taking random values 10^8 times or we can take 10^8 random values for x,y,z all at once and then consider them as the successive displacements. In either case, the result is a reasonable approximation of the motion of a photon over one second.

The problem with this approach, is that it takes more than 1 second of computational time to model the motion of a single photon over a single second. Since the photon takes quite a bit longer than one second to exit the sun, it is not advantageous to model it in such a way. Instead we find the average displacement of a photon over one second along with its standard deviation. The average displacement, of course, is zero, since the photon is equally likely to move in any direction. The standard deviation along any cardinal direction, however, is found to be approximately 57.732 meters.

2.2 Motion over longer periods

Using this value we can then model the motion of a photon over longer time intervals, by expressing the photon's displacement in one second as a random number from the gaussian distribution centered on zero with standard deviation of 57.732. In the instructions for this project we were told to model the motion over an entire year using these values, but as it was highly time-consuming (1 year = 31536000 seconds, and to model it for a sufficiently high number of particles (N=10000 or more) would take nearly 2 hours on my machine), I decided to break the process down further by modelling the photon's motion over 86400 seconds (1 day), then use the resultant standard deviation to model the photon's motion over the entire year (365 days). At this point it was difficult to run the simulation for large number of particles, but it was found that the standard deviation of the displacement of a photon over a year is approximately 320000 meters.

2.3 Motion of a photon as it exits the sun

At any given time, the average displacement of a photon undergoing a random walk will be zero. So if we try to find out when the photon's position will be outside the sun's radius, we will get no satisfactory answer. However, as the number of scatterings increase, the standard deviation of the average value of displacement also increases. Indeed, the standard deviation increases linearly with the square root of the number of scatterings.

Therefore a reasonable way to determine when a photon will have moved from the center of the sun to outside its photosphere is to take the time when the standard deviation of its position is larger than the radius of the sun. of course, this is a probabilistic value, since the photon is only likely to be at that particular distance from the center, and when running multiple simulations we will get multiple values for the time taken to get that value of standard deviation. Thus the simulation must be run a sufficiently large number of times to achieve a high enough degree of confidence in the average time taken for a photon to move from the center of the sun to outside its photosphere (i.e. outside the sun).

3 Results and Analysis

3.1 Simulating photon random walk over one second

We ran our code in "test_ one particle.py" to perform 10^8 random walk steps for a photon, representing approximately 1 second of the photon's journey. This took 8.6875704 seconds to complete, which is significantly more than the time it is meant to simulate.

When we used the provided photon distances data ("photonDistances2.dat") for 10000 photons, ("test_10000_photons.py")we found that the rms distance and the standard de-

viation match closely to the predicted values. A histogram of the data showed close agreement with the gaussian shape, but the peak was skewed slightly to the left.

3.2 Simulating photon random walk over one year

Using the standard deviation derived from our 1s simulation, we could now simulate the random walk for an entire year. I found that to simulate the random walk of a single photon over 1 year (31536000 seconds) took my machine about 3.7531 seconds. This is not an unreasonable time on its own, but if I were to attempt to simulate the motion of the photon over 4.6 million years, this would take nearly 200 days to complete. Therefore we attempted to perform the same trick as the last segment, to find the standard deviation of the photon's displacement over one year.

At this point, I encountered a problem: while the motion of a single photon was quick enough to simulate, to get a reliable value for the standard deviation would require simulating 10000 or more photons, and this was taking far too long on my machine. So I decided to break down the process by adding an intermediary step - instead of simulating the entire year for 10000 photons, I simulated the random walk of 10000 photons for 1 day using the code in "photon-one-day.py". This gave me the standard deviation of 16900 meters, which I then used to simulate the random walk of 10000 photons over 365 days (i.e. 1 year) using the code in "photon-one-year.py". This gave me the final standard deviation of approximately 320000 meters.

3.3 Determining average time for random walk to exit the sun

The radius of the sun is approximately 696000 kilometers, or 6.96×10^8 meters. This is significantly bigger than the values we found for a single year of random walk for a photon. Therefore the simulation must be run for a longer period of time to determine the average time needed for the photon to exit the photosphere of the sun. This was done by running the code in "photon-one-year-step.py", with each step of the simulation representing the displacement of a photon undergoing random walk over one year. It was found that the average time taken for the standard deviation of the photon's displacement to grow larger than the radius of the sun was about 4.5 million years.

The code in "photon-random-walk.py" can be used to generate three text files containing data for the coordinates of a photon undergoing random walk for 1 second with single-collision steps, for 1 year with 1 second steps, and for 4.5 million years with 1 year steps, respectively. As instructed, we plotted the scatter paths from each segment using three plots (XY, YZ, XZ planes) using the code in "plot-random-walk.py".

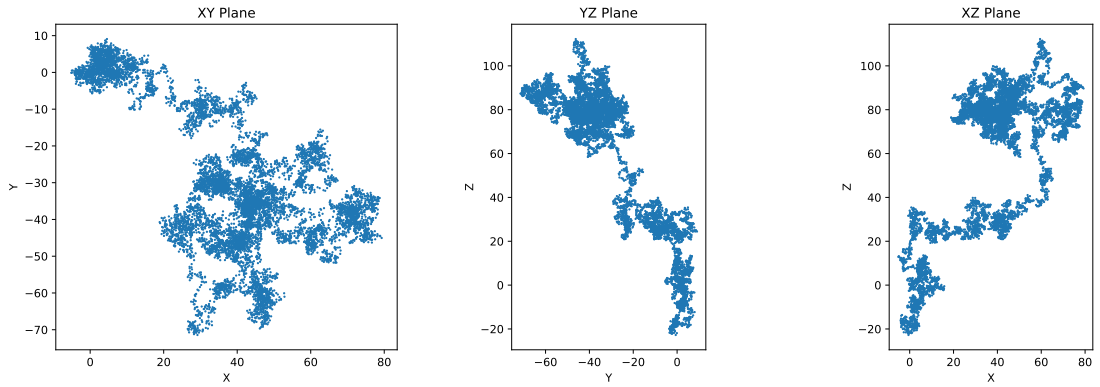


Figure 1: Plot of scattering path for a photon with 10^8 steps per second

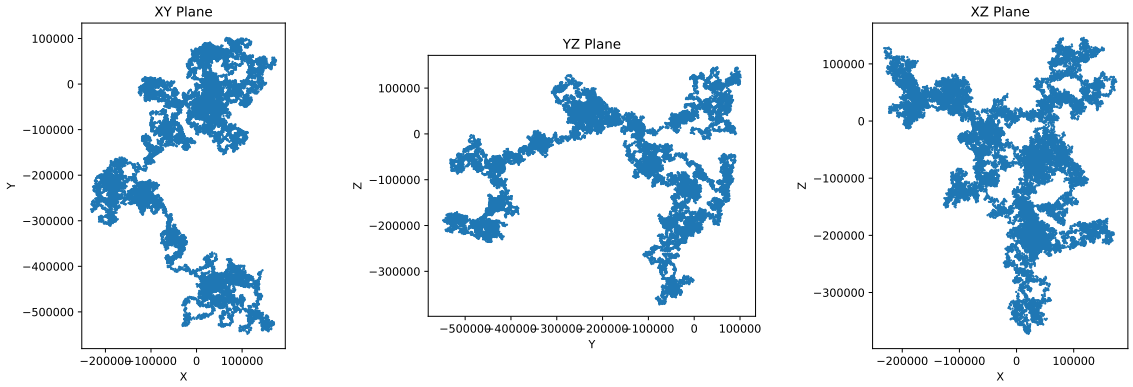


Figure 2: Plot of scattering path for a photon with 1 step per second over 1 year

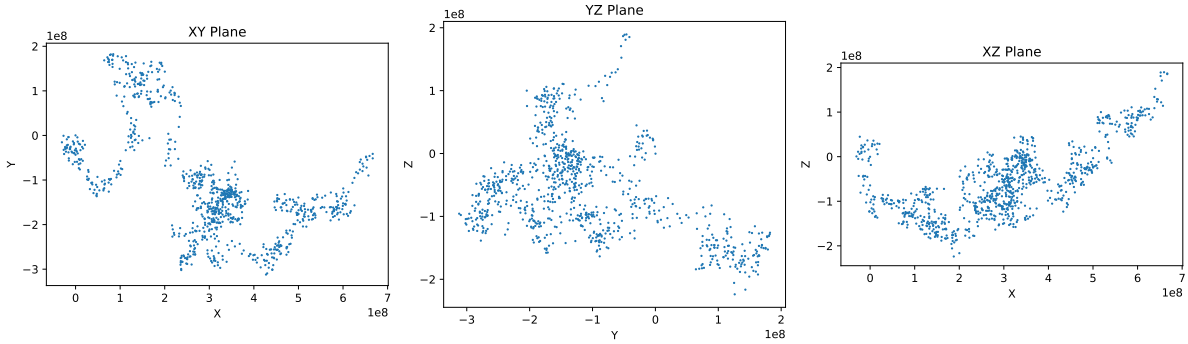


Figure 3: Plot of scattering path for a photon with 1 step per year until it exits the sun

The plots demonstrate the random nature of the scattering quite clearly.

4 Discussion

We found that the simulation of the random walk of a photon using all the scatterings in a single second took 8.6875704 seconds. Ultimately, we determined that the average time taken for a photon to escape the sun's photosphere is 4.5 million years. If we were

to attempts to simulate all the scatterings over this entire period, the estimated total computation time would be

$$T = 8.6875704 \times 4.5 \times 10^6 years = 39.094 \times 10^6 years \tag{2}$$

This is certainly not a reasonable time to invest in a simulation!

One way that we could solve the problem of how long it will take a computer to run a certain code is parallelization. The code that we have been running is what is a serial code, where the program waits for one instruction to complete before performing the next. However, since all the scatterings here are independent from each other, we can compute the change in position due to a scattering in one thread, while computing the change in position due to another scattering in another thread. Modern CPUs (central processor units) are capable of multithreading, or parallel execution. Assuming each of these threads can simulate the effects of a single scattering in the exact same time, to complete the program in 24 hours would require approximately $\frac{39.094\times10^6\times31536000}{24\times3600} = 1.427 \times 10^{10}$ threads. Even with access to a supercomputer with nodes having two processors with 128 threads each, we would need an enormous number of nodes to complete such a simulation within the given time frame of 24 hours. The number of nodes needed can be calculated:

$$N = \frac{1.427 \times 10^{10}}{2 \times 128} = 5.574 \times 10^7 \tag{3}$$

Of course, this was a very basic simulation. One of the key assumptions was that the scattering motion of the photon remains identical at all times as it moves from the center of the sun to the surface. However, this is not the case in reality. The composition, density and temperature of the plasma making up the sun varies with distance from the center. There are also large-scale structures within the sun that exert significant changes in local electromagnetic fields. All of these factors can influence the mean free path of a photon undergoing random walk. Determining the exact changes in the mean free path is difficult, but if we know how the mean free path will be changed in different regions of the sun, then we can adjust our simulation to account for it.

If the mean free path of the photon is known in any given region, then we can correlate the mean free path to the spatial coordinates of the photon's position. This can be done using either cartesian or spherical coordinates, as appropriate. A conditional statement can be added to check the position of the photon at each step, then apply the appropriate mean free path to the step. Then it can be further streamlined based on the exact relations between the coordinates and the factors influencing the mean free path.

5 Conclusion

A point of interest concerning random walks is the statistical nature of the results. Much of our algorithm relied on the standard deviation, and the final result was itself an average value with a rather large standard deviation of its own. Thus it is less a prediction of how long it will take a photon to exit the sun than it is an estimate of what fraction of photons generated in the center of the sun reach the surface over time.

The direct simulation requires far too much time to be practical, however, and even then the element of randomness remains, so statistical techniques must be utilized to improve computation efficiency.